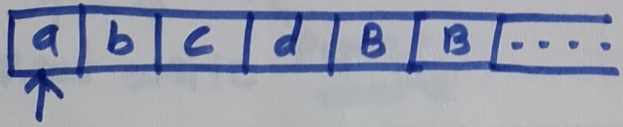
MODULE - 05

TURING MACHINE (TM)

What is tuzing Machine

- TM is an accepting machine which accepts recursively enumerable language generated by type 0 grammar.
- It was invented by Alan Turing in 1936.
- It is a mathematical model which consist of an infinite length & tape is divided into cells on which I/P is given
- It consist of head which reads the I/P tape
- After reading an I/P symbol, it is replaced with another symbol, its internal state is changed & it moves from one cell to the right or left
- If TM reaches the final state, the I/P string is accepted



Representation of TM

M is represented by 7 tuples

$$M = (Q, \Sigma, \delta, q_0, B, \Gamma, F)$$

where

$Q \rightarrow$ Finite set of states

$\Sigma \rightarrow$ I/P alphabets ($\neq B$)

$q_0 \rightarrow$ Initial state

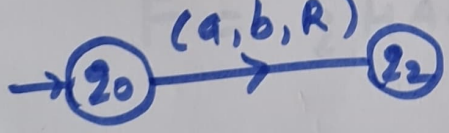
$B \rightarrow$ Blank symbol

$\Gamma \rightarrow$ Tape alphabet (B can be considered)

$F \rightarrow$ Final state

$\delta \rightarrow$ Transition function

$$\delta : Q \times \Sigma \rightarrow Q \times \Gamma \times [\text{left shift (L)}, \text{Right Shift (R)}]$$

eg 

$$\delta(q_0, a) = (q_2, b, R)$$

(a, b, R)

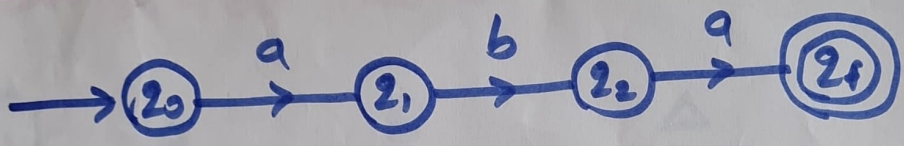
↑ Read ↑ write ↑ Right move

Construct a TM which accepts the language aba over $\Sigma = \{a, b\}$

⇒ **Step 1** Theory –

To design TM to accept the language aba over $\Sigma = \{a, b\}$

Step 2 Logic, tuple format



$M = (Q, \Sigma, \delta, q_0, B, \Gamma, F)$

$Q = \{q_0, q_1, q_2, q_3, \text{HALT}\}$

$\Sigma = \{a, b\}$

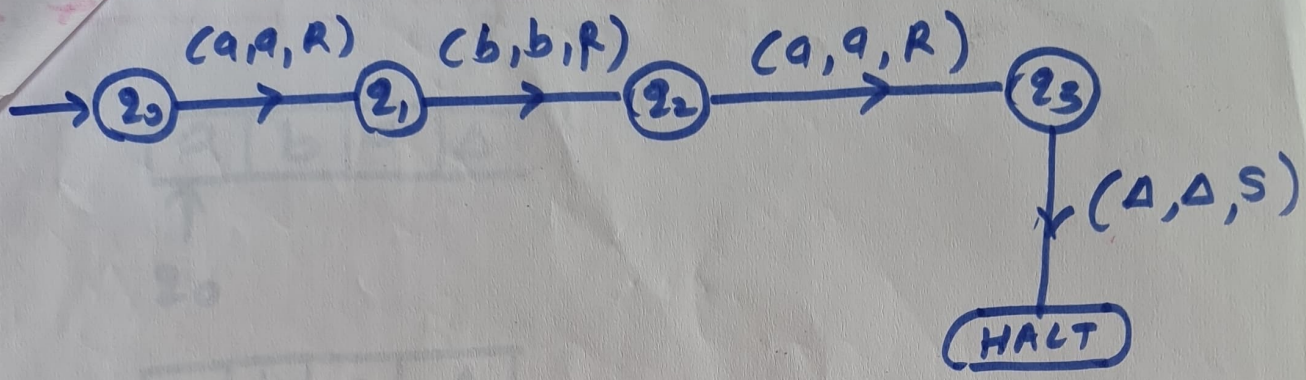
$q_0 = q_0$

$\Gamma = \{a, b, \Delta\}$

$B = \Delta$

$F = \{\text{HALT}\}$

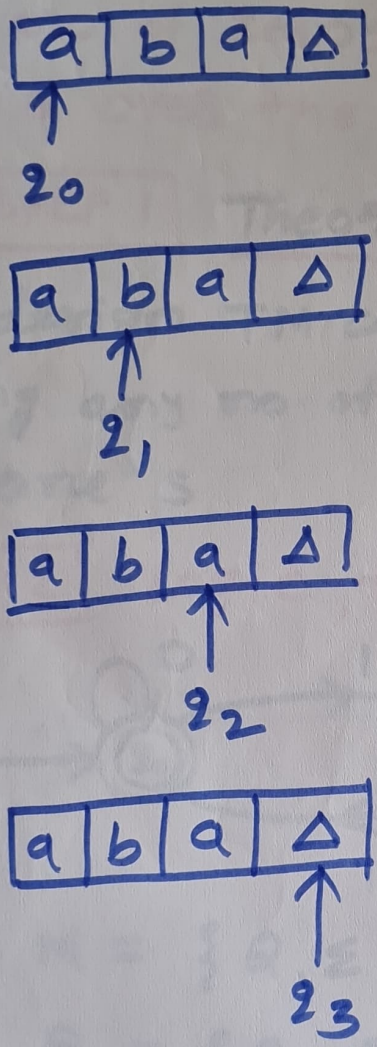
Step 3 Implementation



Transition Table

q \ Γ	a	b	Δ
→ q_0	(q_1, a, R)	–	–
q_1	–	(q_2, b, R)	–
q_2	(q_3, a, R)	–	–
q_3	–	–	(HALT, Δ, S)

Step 4 Example

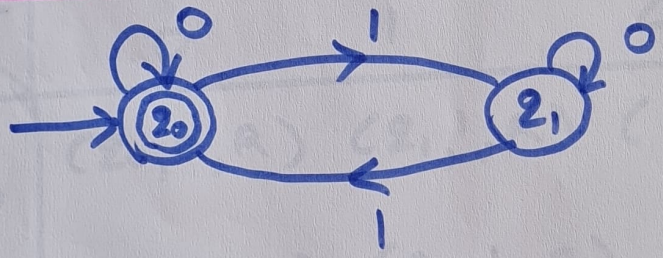


Construct turing m/c for the language consisting of a string having any no. of zeros & only even no. of one's over the i/p alphabet $\Sigma = \{0, 1\}$

⇒ **Step 1** **Theory**

To design TM which accepts string having any no. of zeros & even no. of one's

Step 2 Logic, tuple format



$M = \{Q, \Sigma, \delta, q_0, B, \Gamma, F\}$

$Q = \{q_0, q_1, \text{HALT}\}$

$\Sigma = \{0, 1\}$

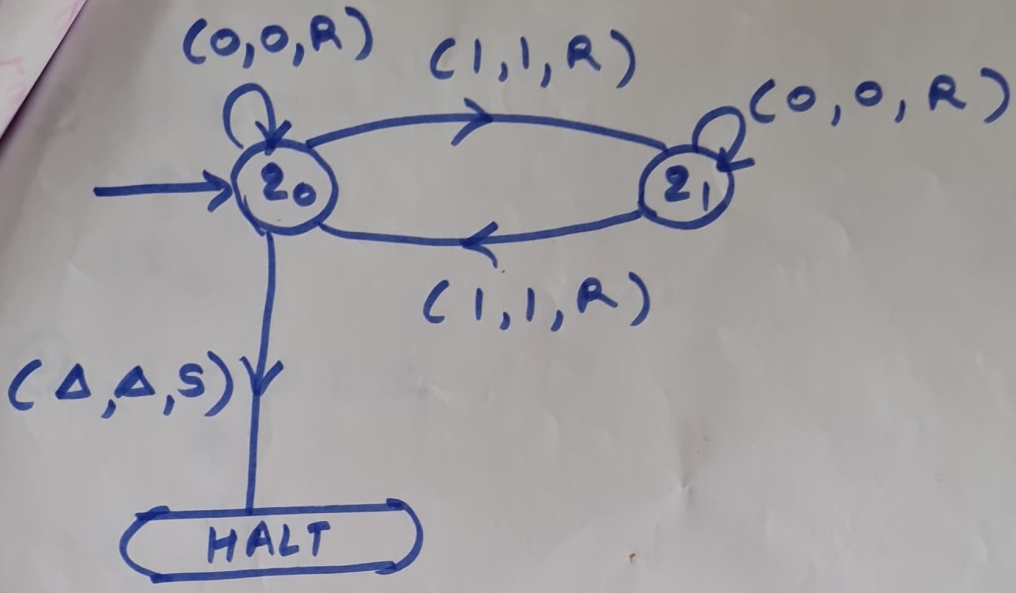
$q_0 = q_0$

$\Gamma = \{0, 1, \Delta\}$

$B = \Delta$

$F = \{\text{HALT}\}$

EP 3 Implementation



Transition Table

Q \ Γ	0	1	Δ
→ 2_0	$(2_0, 0, R)$	$(2_1, 1, R)$	(HALT, Δ, S)
2_1	$(2_1, 0, R)$	$(2_0, 1, R)$	—

STEP 4 Example

① 1 0 0 1 Δ
↑
z_0

② 1 0 0 1 Δ
↑
z_1

③ 1 0 0 1 Δ
↑
z_1

④ 1 0 0 1 Δ
↑
z_0

⑤ 1 0 0 1 Δ
↑
z_0 HALT

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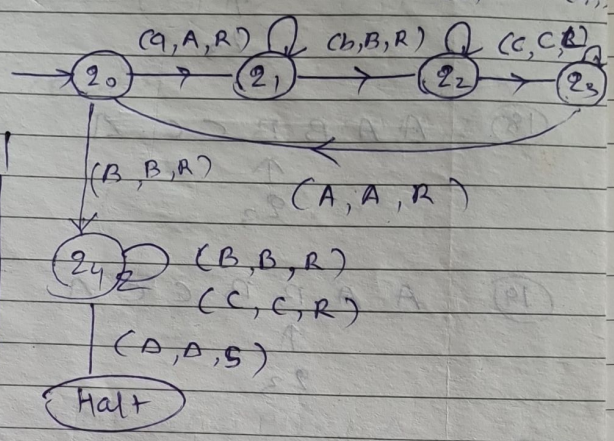
$L = \{a^n b^n c^n\}$

aabbccΔ



aabbccΔ
↑
q_0

AabbccΔ
↑
q_1

AabbccΔ
↑
q_1

AaBbccΔ
↑
q_2

AaBbCcΔ
↑
q_2

~~AaBbCcΔ~~
~~↑~~
~~q_3~~

AaBbCcΔ
↑
q_3

AaBbCcΔ
↑
q_3

AaBbCcΔ
↑
q_3

(10) AaBbCcΔ
↑
q_3

(11) AaBbCcΔ
↑
q_3

(12) AABbCcΔ
↑
q_0

(13) AABbCcΔ
↑
q_1

(14) AABBCcΔ
↑
q_2

(15) AABBCcΔ
↑
q_2

(16) AABBCcΔ
↑
q_3

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$L = a^n b^n$

(17) A A B B C C Δ
↑ q_3

(18) A A B B C C Δ
↑ q_3

(19) A A B B C C Δ
↑ q_3

(20) A A B B C C Δ
↑ q_3

(21) A A B B B C C Δ (Before B there A)
↑ q_0

(22) A A B B C C Δ
↑ q_4

(23) A A B B C C Δ
↑ q_4

(24) A A B B C C Δ
↑ q_4

(25) A A B B C C Δ
↑ q_4

a a a b b b Δ
↑ q_0

A a a b b b Δ
↑ q_1

A a a b b b
↑ q_1

A a a b b b
↑ q_1

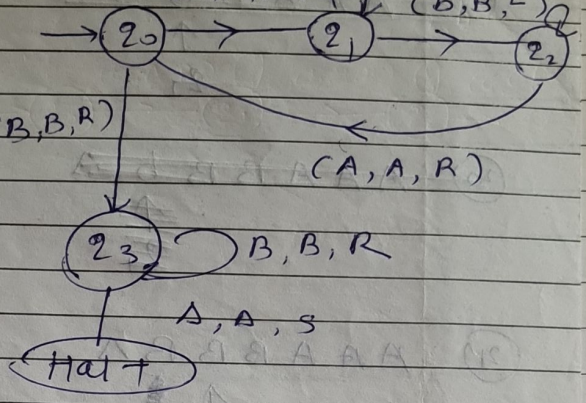
A a a
↑ q_2

A a a
↑ q_2

A a
↑ q_2

A

$L = a^n b^n$



q_0 a a a b b b Δ

A q_1 a a b b b Δ

A a q_1 a b b b Δ

A a a q_1 b b b Δ

A a a B q_2 b b Δ

A a q_2 a B b b b Δ

A q_2 a a B b b b Δ

A a a B b b b Δ
q_2

A a a B b b b Δ
q_0

A A a B b b b Δ
q_1

A A a B b b b Δ
q_1

A A a B b b b Δ
q_0

A A a B b b b Δ
q_1

A A a B B b b Δ
q_2

A A a B B b Δ
q_2

A A a B B b Δ
q_2

A A a B B b Δ
q_2

A A a B B b Δ
q_0

A A A B B b Δ
q_1

A A A B B b Δ
q_1

A A A B B b Δ
q_1

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(19) A A A B B b Δ
↑
q_1

(20) A A A B B b Δ
↑
q_1

(21) A A A B B B Δ
↑
q_2

(22) A A A B B B Δ
↑
q_2

(23) A A A B B B Δ
↑
q_2

(1) A A A B B B Δ
↑
q_2

Now immediately before B is A that means all the a's are marked by A. So we will move right to ensure that no b is present.

A A A B B B Δ
↑
q_3

A A A B B B Δ
↑
q_3

A A A B B B Δ
↑
q_3
Stop

q_0

Design a TM
= 01*0

(0, 0, R)

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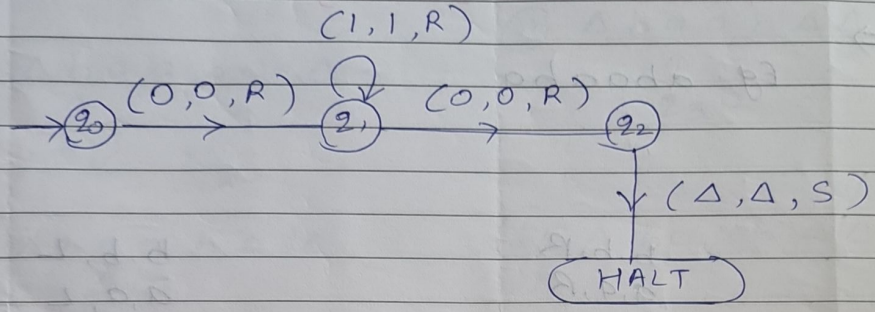
Design a TM which recognizes the $L = 01^*0$



7 - tuple format

$M = (Q, \Sigma, \Gamma, q_0, \delta, B, F)$

$Q = \{q_0, q_1, q_2, HALT\}$
$\Sigma = \{0, 1\}$
$\Gamma = \{0, 1, \Delta\}$
$q_0 = q_0$
$B = \Delta$
$F = HALT$

① 0 1 1 0 Δ
↑
q_0

② 0 1 1 0 Δ
↑
q_1

③ 0 1 1 0 Δ
↑
q_1

④ 0 1 1 0 Δ
↑
q_1

⑤ 0 1 1 0 Δ
↑
q_2 HALT

⑥ ~~0 1 1 0 Δ~~

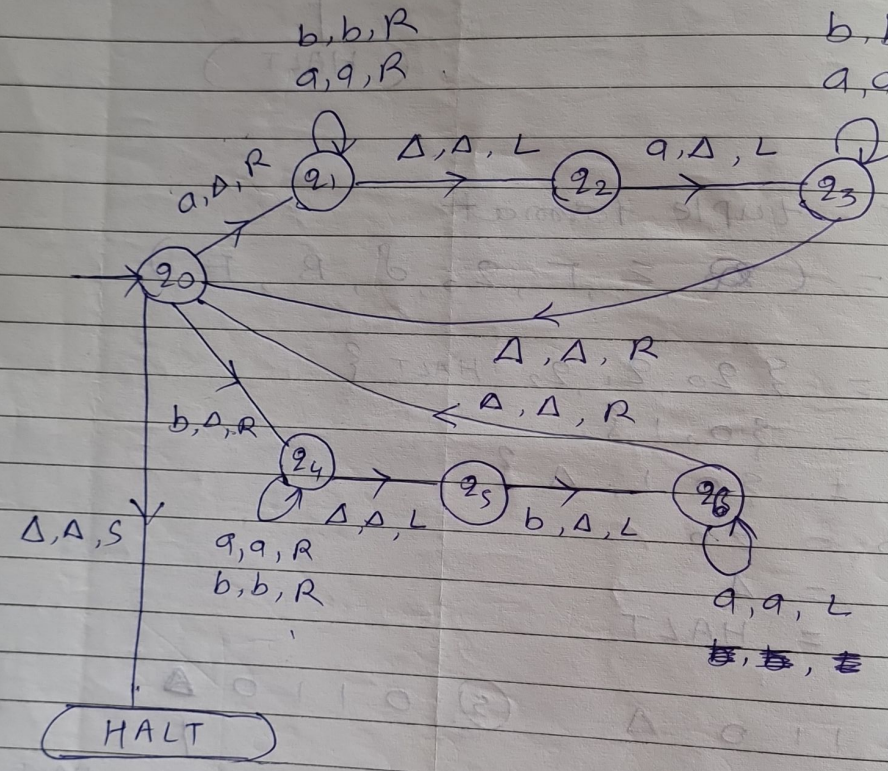
Transition Table

Q \ Γ	0	1	Δ
→ q_0	$(q_1, 0, R)$	~~$(q_1, 1, R)$~~	–
q_1	$(q_2, 0, R)$	$(q_1, 1, R)$	–
q_2	–	–	(HALT, Δ, S)

Design a turing machine that accepts even palindrome ones the alphabet $\Sigma = \{a, b\}$

$\Rightarrow$

Eg abaaba



$M = (Q_0, \Sigma, \delta, \Gamma, B, Q, F)$

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, HALT\}$

$\Sigma = \{a, b\}$

$\Gamma = \{a, b, \Delta\}$

$B = \{\Delta\}$

$q_0 = q_0$

$F = HALT$

the addition

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a b a a b a Δ
↑
q_0

Δ b a a b a Δ
↑
q_1

(3) Δ b a a b a Δ
↑
q_1

(4) Δ b a a b a Δ
↑
q_1

(5) Δ b a a b a Δ
↑
q_1

(6) Δ b a a b a Δ
↑
q_1

(7) Δ b a a b a Δ
↑
q_1

(8) Δ b a a b a Δ
↑
q_2

(9) Δ b a a b Δ Δ
↑
q_3

(10) Δ b a a b Δ Δ
↑
q_3

(11) Δ b a a b Δ Δ
↑
q_3

(12) Δ b a a b Δ Δ
↑
q_3

(13) Δ b a a b Δ Δ
↑
q_3

(14) Δ b a a b Δ Δ
↑
q_0

(15) Δ Δ a a b Δ Δ
↑
q_4

(16) Δ Δ a a b Δ Δ
↑
q_4

(17) Δ Δ a a b Δ Δ
↑
q_4

(18) Δ Δ a a b Δ Δ
↑
q_4

(19) Δ Δ q a b Δ Δ
↑
q_5

(20) Δ Δ q a Δ Δ
↑
q_6

(21) Δ Δ q a Δ Δ Δ
↑
q_6

(22) Δ Δ q a Δ Δ Δ
↑
q_6

(23) Δ Δ q a Δ Δ Δ
↑
q_0

(24) Δ Δ Δ q Δ Δ Δ
↑
q_1

(25) Δ Δ Δ q Δ Δ Δ
↑
q_1

(26) Δ Δ Δ q Δ Δ Δ
↑
q_2

(27) Δ Δ Δ Δ Δ Δ Δ
↑
q_3

(28) Δ Δ Δ Δ Δ Δ Δ
↑
q_0 HALT

(29) Transition Table

Q \ Γ	a	b	Δ
→ q_0	(q_1, Δ, R)	(q_4, Δ, R)	(HALT, Δ, S)
q_1	(q_1, a, R)	(q_1, b, R)	(q_2, Δ, L)
q_2	(q_3, Δ, L)	—	—
q_3	(q_3, a, L)	(q_3, b, L)	(q_0, Δ, R)
q_4	(q_4, a, R)	(q_4, b, R)	(q_5, Δ, L)
q_5	(q_5, a, L)	~~(q_5, b, L)~~	~~(q_0, Δ, R)~~
q_6	(q_6, a, L)	~~(q_6, b, L)~~	(q_0, Δ, R)
HALT			

Construct T
function f
system.

⇒ The wo
of one
can be
as l
For t
add
eg

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* Construct TM for the addition function for the unary number system.

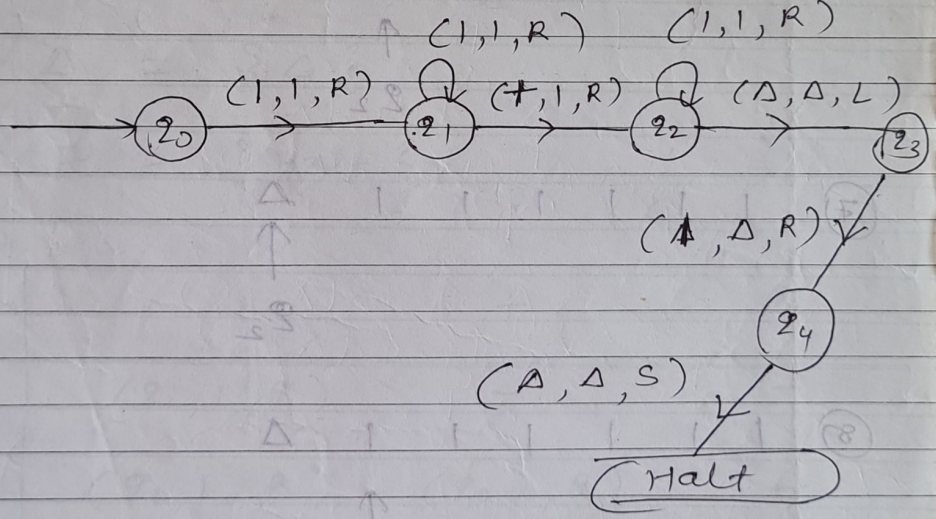
⇒

- The unary no. system is made up of only one character i.e. no. 5 can be written in unary no. system as 11111
- In this TM we are going to perform addition of 2 unary no's.

eg 2 + 3

11 + 111 = 11111

- Simply replace + by 1 & move ahead right for searching end of the string. We will convert last 1 to Δ.



① 1 1 1 + 1 1 Δ

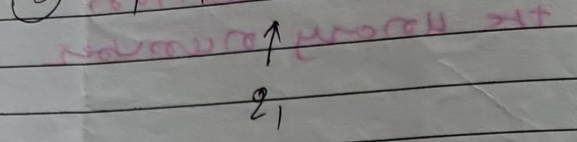
↑

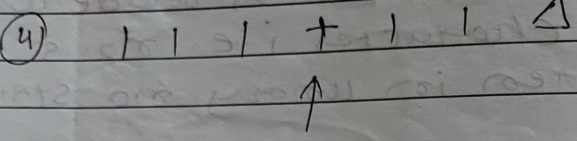
q_0

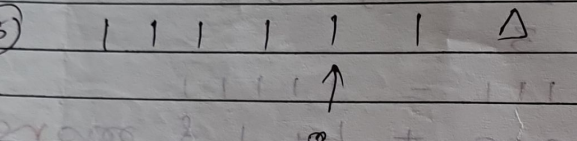
② 1 1 1 + 1 1 Δ

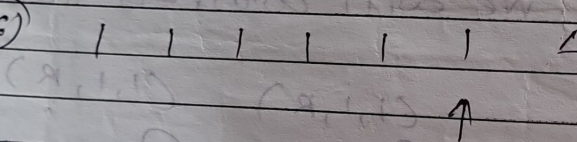
↑

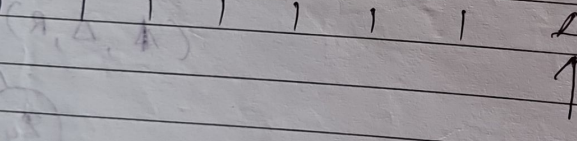
q_1

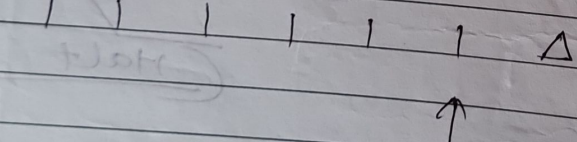
(3) 

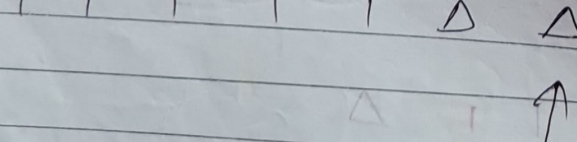
(4) 

(5) 

(6) 

(7) 

(8) 

(9) 

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machine

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2) 1 1 1 1 1 Δ Δ

stop

$M = (Q, \Sigma, q_0, \delta, \Gamma, \Delta, F)$

$Q = \{q_0, q_1, q_2, q_3, q_4, Halt\}$

$\Sigma = \{1\}$

$q_0 = \{q_0\}$

$\delta =$

$\Gamma = \{1, +, \Delta\}$

$\Delta = \{\Delta\}$

$F = \{Halt\}$

	1	+	Δ
q_0	$(q_1, 1, R)$	–	–
q_1	$(q_1, 1, R)$	$(q_2, 1, R)$	–
q_2	$(q_2, 1, R)$	–	(q_3, Δ, L)
q_3	(q_4, Δ, R)	–	–
q_4	–	–	$(Halt, \Delta, S)$
Halt			

B
↑
q_1
1 B
↑
q_2
1 B B
↑
q_3
0 1 B
↑
q_3

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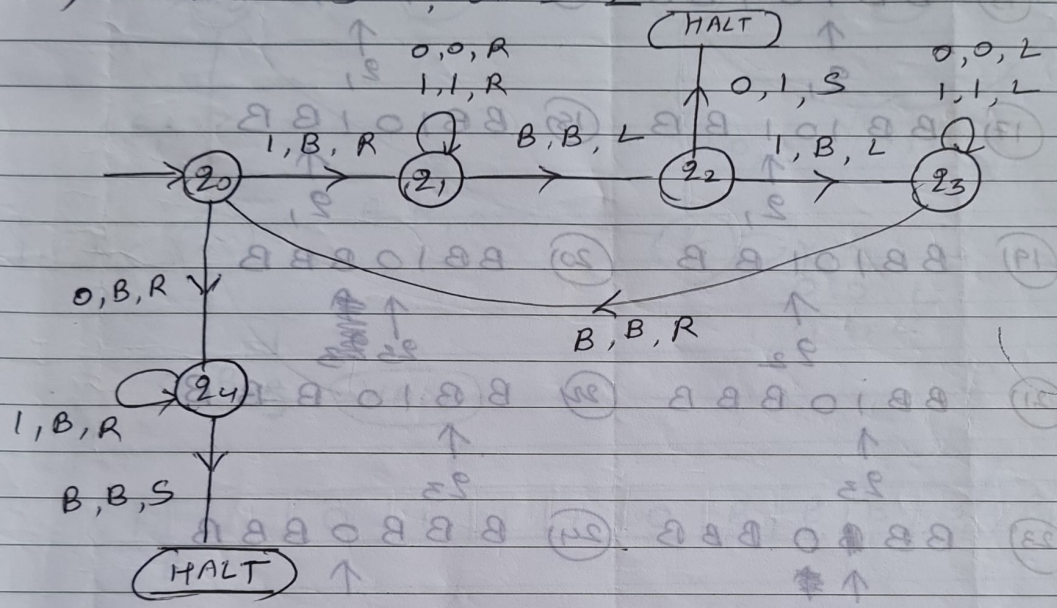
Construct a turing machine that perform subtraction function

$$f(m,n) = \begin{cases} m-n, & m \geq n \\ 0, & m \leq n \end{cases}$$

$\Rightarrow$ $m = 3, n = 2$



① B 1 1 1 0 1 1 B
↑ q_0

② B B 1 1 0 1 1 B
↑ q_1

③ B B 1 1 0 1 1 B
↑ q_1

④ B B 1 1 0 1 1 B
↑ q_1

⑤ B B 1 1 0 1 1 B
↑ q_1

⑥ B B 1 1 0 1 1 B
↑ q_1

⑦ B B 1 1 0 1 1 B
↑ q_1

⑧ B B 1 1 0 1 1 B
↑ q_2

⑨ B B 1 1 0 1 B B
↑ q_3

⑩ B B 1 1 0 1 B B
↑ q_3

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(11) B 1 1 0 1 B B
↑ q_3

(12) B 1 1 0 1 B B
↑ q_3

(13) B 1 1 0 1 B B
↑ q_3

(14) B 1 1 0 1 B B
↑ q_0

(15) B B 1 0 1 B B
↑ q_1

(16) B B 1 0 1 B B
↑ q_1

(17) B B 1 0 1 B B
↑ q_1

(18) B B 1 0 1 B B
↑ q_1

(19) B B 1 0 1 B B
↑ q_2

(20) B B 1 0 B B B
↑ q_3

(21) B B 1 0 B B B
↑ q_3

(22) B B 1 0 B B B
↑ q_3

(23) B B B 0 B B B
↑ q_0

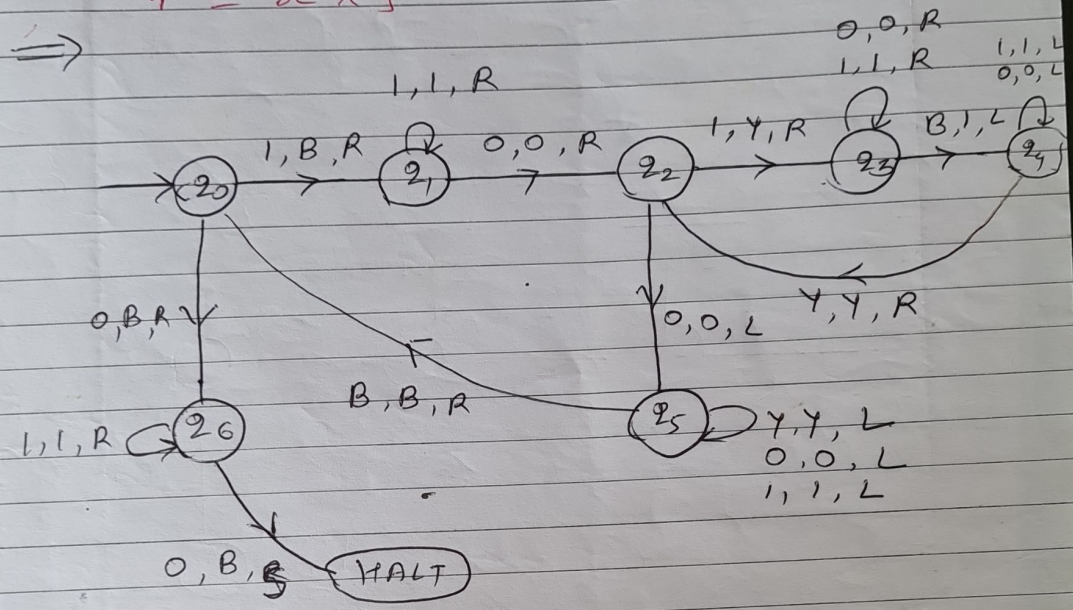
(24) B B B 0 B B B
↑ q_1

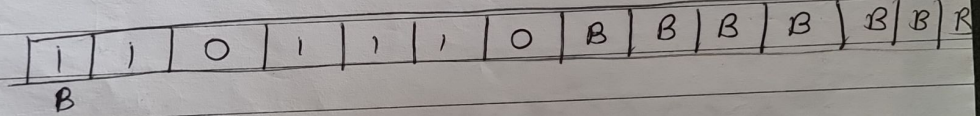
(25) B B B 0 B B B
↑ q_1

(26) B B B 0 B B B
↑ q_2

(27) B B B 1 B B B
↑ HALT

Construction of a turing machine for multiplication operation
$f = x \times y$





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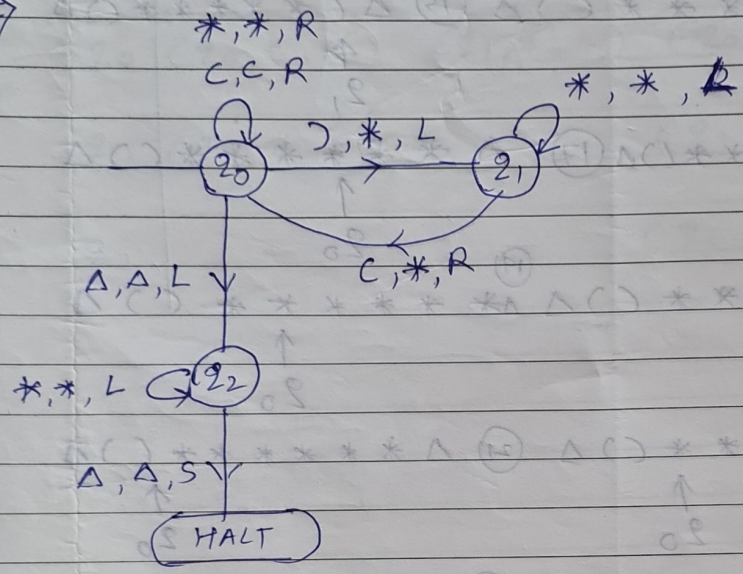
Design turing machine to check well formness of Paranthesis (well formed Paranthesis)

⇒



① Δ (( ) ( )) ( ) Δ
↑ q0

② Δ (( ) ( )) ( ) Δ
↑ q0

③ Δ (( ) ( )) ( ) Δ
↑ q0

④ Δ ((* ( )) ( ) Δ
↑ q1

⑤ Δ * * * ( )) ( ) Δ
↑ q0

⑥ Δ (* * ( )) ( ) Δ
↑ q0

⑦ Δ (* * ( )) ( ) Δ
↑ q0

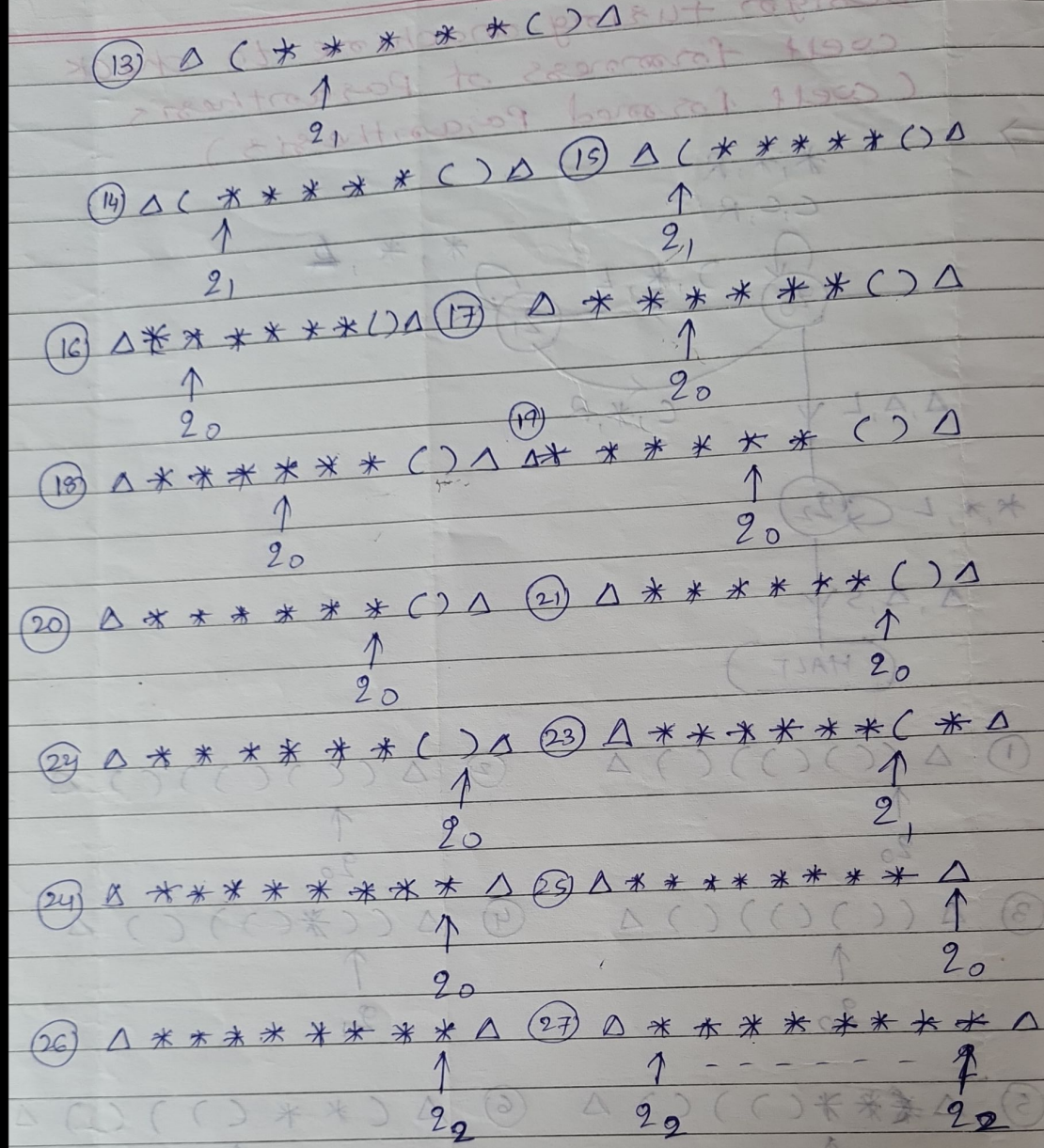
⑧ Δ (* * (*) ( ) Δ
↑ q1

⑨ Δ (* * * *) ( ) Δ
↑ q0

⑩ Δ (* * * *) ( ) Δ
↑ q0

⑪ Δ (* * * * * ( ) Δ
↑ q1

⑫ Δ (* * * * * ( ) Δ
↑ q1



Implementation

$M = (Q, q_0, \Sigma, \delta, F, \Gamma, B)$

$Q = (q_0, q_1, q_2)$

$q_0 = q_0$

$\Sigma = \{ (,) \}$

$F = HALT$

$\Gamma = \{ (,), *, \Delta \}$

$B = \{ \Delta \}$

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Transition Table -

Γ	(	)	*	Δ
→ q_0	(q_0, c, R)	$(q_1, *, L)$	$(q_0, *, R)$	–
q_1	$(q_0, *, R)$	–	$(q_1, *, R)$	–
q_2	–	–	$(q_2, *, L)$	(HALT, Δ, S)

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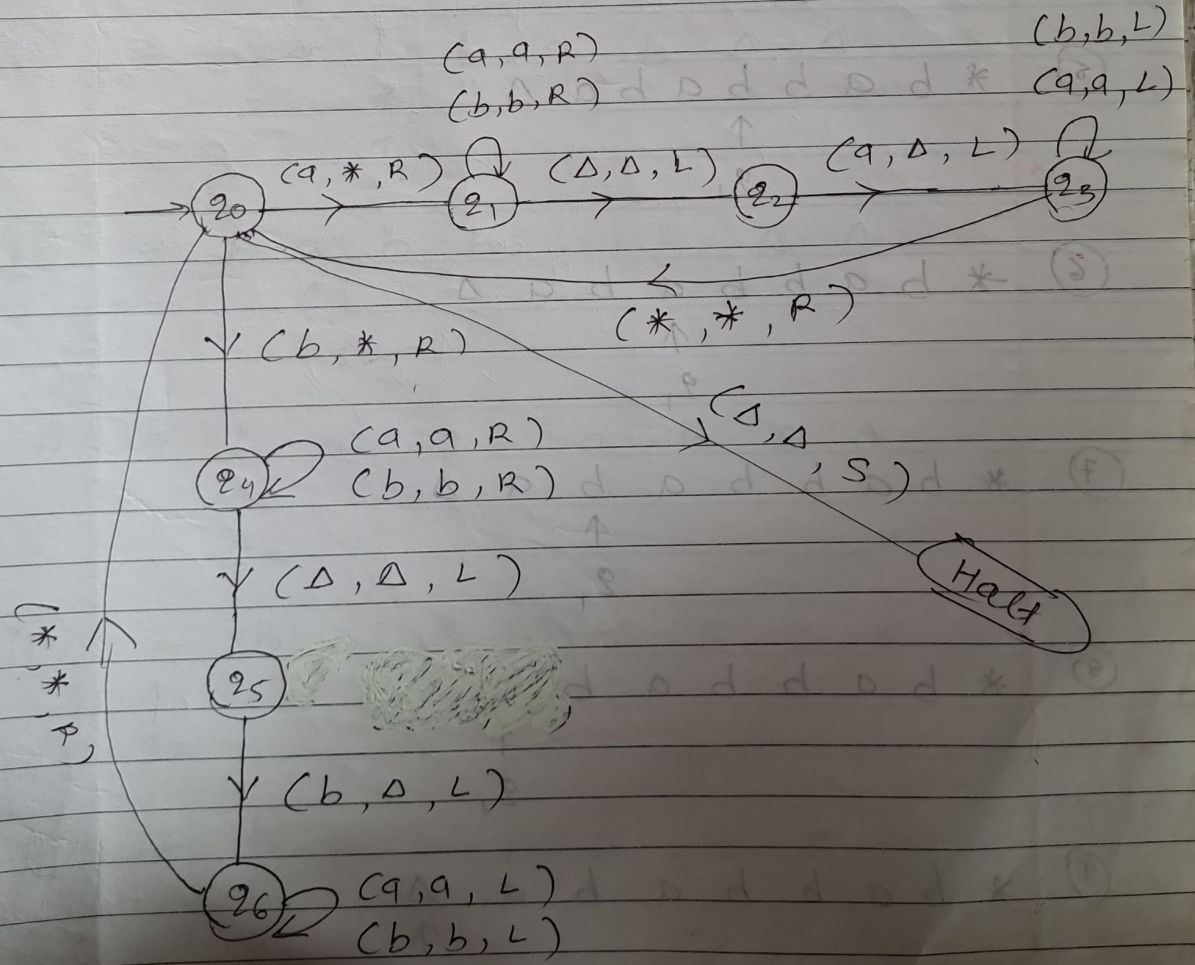
* Construct a TM for checking the palindrome of the string of even length.

⇒ The logic is that we will read first symbol from left & then we compare it with the first symbol from right to check whether it is the same
- Again we will compare second symbol from left with the second symbol from right.
- We repeat this process for all the symbols.
- If we found any symbol not matching we can not lead the m/c to halt state.



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① a b a b b a b a Δ
↑
q_0

② * b a b b a b a Δ
↑
q_0

③ * b a b b a b a Δ
↑
q_1

④ * b a b b a b a Δ
↑
q_1

⑤ * b a b b a b a Δ
↑
q_1

⑥ * b a b b a b a Δ
↑
q_1

⑦ * b a b b a b a Δ
↑
q_1

⑧ * b a b b a b a Δ
↑
q_1

⑨ * b a b b a b a Δ
↑
q_1

⑩ * b

⑪ * b

⑫ * b

⑬ * b

⑭ * b

⑮ * b

⑯ *

⑰ *
↑
q_3

⑱ *

(10) * b a b b a b a Δ

q_2

(11) * b a b b a b Δ Δ

q_3

(12) * b a b b a b Δ Δ

q_3

(13) * b a b b a b Δ Δ

q_3

(14) * b a b b a b Δ Δ

q_3

(15) * b a b b a b Δ Δ

q_3

(16) * b a b b a b Δ Δ

q_3

(17) * b a b b a b Δ Δ

q_3

(18) * b a b b a b Δ Δ

q_0

(19) * * a b b a b Δ Δ
↑
q_4

(20) * * a b b a b Δ Δ
↑
q_4

(21) * * a b b a b Δ Δ
↑
q_4

(22) * * a b b a b Δ Δ
↑
q_4

(23) * * a b b a b Δ Δ
↑
q_4

(24) * * a b b a b Δ Δ
↑
q_4

(25) * * a b b a b Δ Δ
↑
q_5

(26) * * a b b a Δ Δ Δ
↑
q_6

(27) * * a b b a Δ Δ Δ
↑
q_6

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(28) * * a b b a △ △ △
↑
q_6

(29) * * a b b a △ △ △
↑
q_6

(30) * * a b b a △ △ △
↑
q_6

(31) * * a b b a △ △ △
↑
q_0

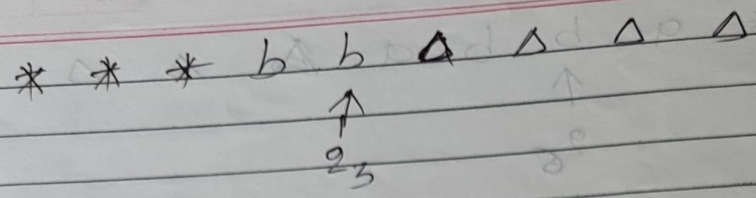
(32) * * * b b a △ △ △
↑
q_1

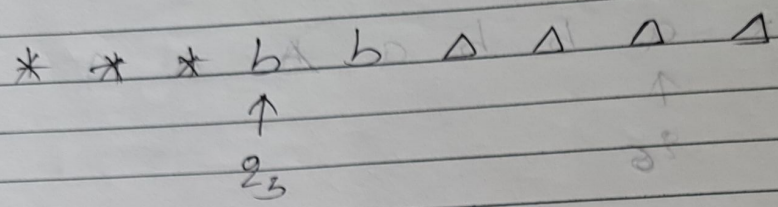
(33) * * * b b a △ △ △
↑
q_1

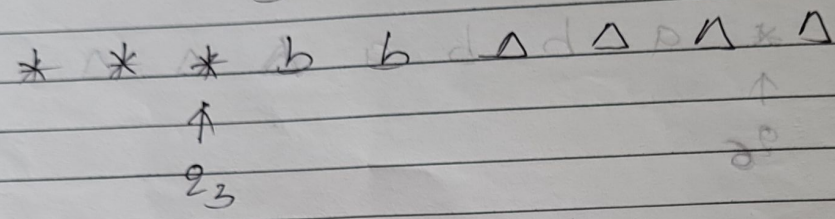
(34) * * * b b a △ △ △
↑
q_1

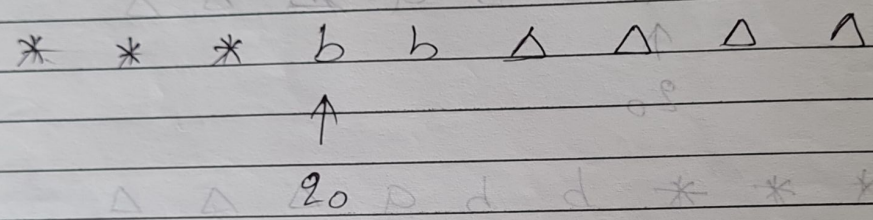
(35) * * * b b a △ △ △
↑
q_1

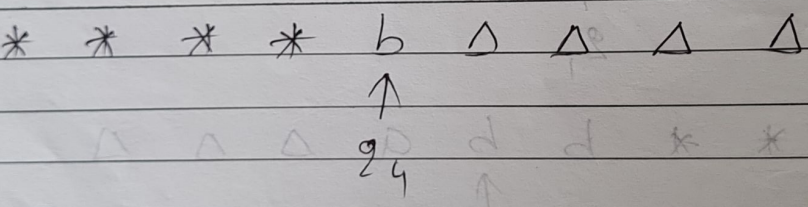
(36) * * * b b a △ △ △
↑
q_2

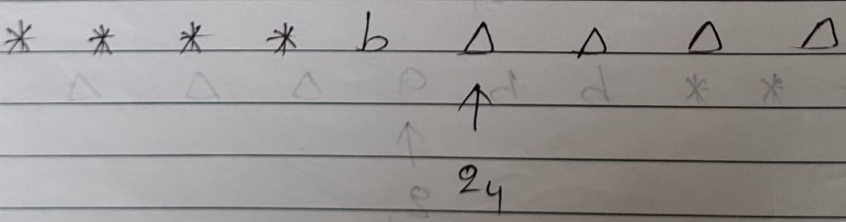
(37) 

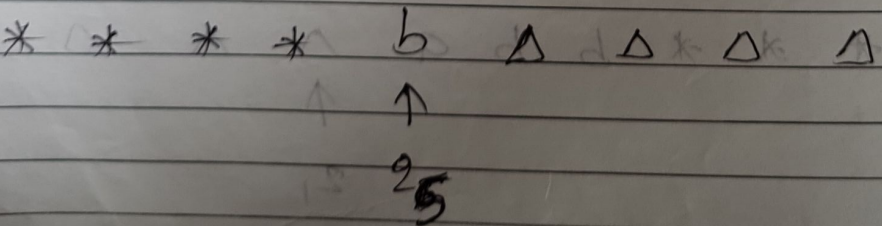
(38) 

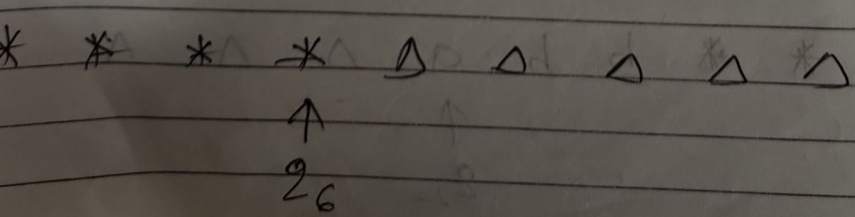
(39) 

(40) 

(41) 

(42) 

(43) 

(44) 

(45) *

T

q_0

q_1

q_2

q_3

q_4

q

(45) * * * * Δ Δ Δ Δ Δ

↑

q_0

Transition Table

	a	b	*	Δ
q_0	$(q_1, *, R)$	$(q_4, *, R)$	–	–
q_1	(q_1, a, R)	(q_1, b, R)	–	(q_2, Δ, L)
q_2	(q_3, Δ, L)	–	–	–
q_3	(q_3, a, L)	(q_3, b, L)	$(q_0, *, R)$	–
q_4	(q_4, a, R)	(q_4, b, R)	–	(q_5, Δ, L)
q_5	–	–	(q_6, Δ, L)	–
q_6	(q_6, a, L)	(q_6, b, L)	$(q_0, *, R)$	–

7 - tuple

$M = (Q, \Sigma, \delta, q_0, \Gamma, \Delta, F)$

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, Halt\}$

$\Sigma = \{a, b\}$

$q_0 = q_0$

$\Gamma = \{a, b, *, \Delta\}$

$\Delta = \Delta$

$F = \{Halt\}$

ALDEL EDUCATION TRUST'S

12

No. 2

St. John College of Engineering and Management

(A Christian Religious Minority Institution)

St. John Technical Campus, Vevoor, Manor Road, Palghar (East)

Dist. Palghar - 401404, Maharashtra, India.

CLASS TEST

Name of the Student : _____ Roll No. _____

Branch/Div. : _____ Semester : _____ Date : _____

Subject : _____ Sign. of Supervisor _____

Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10	Q.11	Q.12	Q.13	TOTAL/MARKS

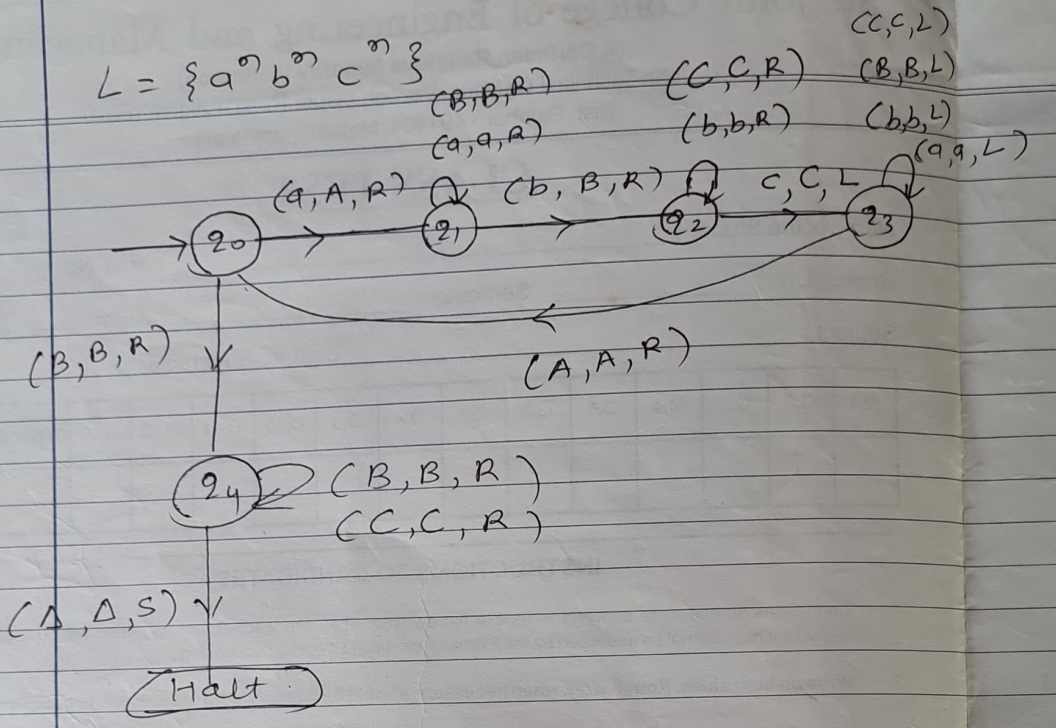
INSTRUCTIONS TO CANDIDATES

1. Candidates should write answers in legible handwriting. They are warned that no marks will be assigned to answers which cannot be assessed by the Examiners owing to illegible handwriting.
2. **Write on both sides. Rough work when necessary, should be done on left-hand side and in pencil only.**
3. No sheet shall be torn from the answer-book provided nor shall any additional paper brought from outside be attached.
4. All answer-books supplied shall be returned, whether written or blank.

START WRITING HERE

A	B	C	Δ
—	(q_4, B, R)	—	—
—	(q_1, B, R)	—	—
—		(q_2, C, R)	—
—	(q_3, B, L)	(q_3, C, L)	—
—	(q_4, B, R)	(q_4, C, R)	(H, Δ, S)

$L = \{a^n b^n c^n\}$



① a a b b c c Δ
↑
q_0

② A a b b c c Δ
↑
q_1

③ A a b b c c Δ
↑
q_1

④ A a B b c c Δ
↑
q_2

⑤ A a B b C c Δ
↑
q_2

⑥ A a B b C c Δ
↑
q_3

⑦ A a B b C c Δ
↑
q_3

⑧ A a B b C c Δ
↑
q_3

⑨ A a B b C c Δ
↑
q_3

⑩ A a B b C c Δ
↑
q_0

⑪ A A B b C c Δ
↑
q_1

⑫ A A B

⑬ A A B

⑭ A A B

⑮ A A B

⑯ A A

⑰ A A

⑱ A A
↑
q_3

⑲ A A

⑳ A A

(12) A A B b C C Δ
↑
q_1

(13) A A B B C C Δ
↑
q_2

(14) A A B B C C Δ
↑
q_2

(15) A A B B C C Δ
↑
q_3

(16) A A B B C C Δ
↑
q_3

(17) A A B B C C Δ
↑
q_3

(18) A A B B C C Δ
↑
q_3

(19) A A B B C C Δ
↑
q_0

Before B there is A

(20) A A B B C C Δ
↑
q_4

(21) A A B B C C Δ
↑
q_4

(22) A A B B C C Δ
↑
q_4

(23) A A B B C C Δ
↑
STOP

δ :

$\delta(q_0, a) = (q_1, A, R)$
$\delta(q_1, b) = (q_2, B, R)$
$\delta(q_1, a) =$

Transition Table

	a	b	Δ
q_0	(q_1, A, R)	—	—
q_1	(q_1, a, R)	(q_2, B, R)	—
q_2	—	(q_2, b, R)	(q_3, C, L)
q_3	(q_3, a, L)	(q_3, b, L)	—
q_4	—	—	—
Halt			